Project Executable Files

Date	1 July 2026
Project Name	OptiCrop
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for the OptiCrop project. All files are organized as listed below and hosted on Render for evaluator access.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	[Fill in: Yes/No/NA]
10	Demo video or walkthrough (if required)	[Fill in: Yes/No/NA]

Step 2: File / Folder Structure

```
opticrop-project/
├── app.py                    - Main Flask application (routes, prediction logic)
├── model/
│   └── crop_recommendation_model.pkl  - Trained Random Forest model
├── templates/
│   ├── index.html            - Home page (Get Started / hero section)
│   ├── about.html            - About OptiCrop section
│   ├── predict.html          - Soil & environmental input form
│   └── result.html           - Crop Recommendation Result page
├── static/
│   ├── css/                  - Stylesheets
│   ├── js/                   - Client-side scripts
│   └── images/               - Hero/background images (wheat field, sunrise, etc.)
├── requirements.txt          - Python dependencies (Flask, scikit-learn, pandas, numpy)
├── render.yaml / Procfile   - Render deployment configuration
└── README.md                 - Setup and run instructions
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://<your-app-name>-onrender.com [complete the full Render URL from your browser address bar]
Login Credentials (Demo)	Not required – the application is publicly accessible without login
Platform / Hosting Provider	Render
Repository Link	[Insert your GitHub repository URL here]

Demo Video Link	[Insert demo video link here, if recorded]

Step 4: Run Instructions

Pre-requisites:

Python 3.9+, pip, and a virtual environment tool (venv/conda). Internet connection for installing dependencies.

Step 1: Clone the repository: git clone <repository-url>

Step 2: Navigate into the project folder: cd opticrop-project

Step 3: (Optional) Create and activate a virtual environment: python -m venv venv && venv\Scripts\activate (Windows) or source venv/bin/activate (Mac/Linux)

Step 4: Install dependencies: pip install -r requirements.txt

Step 5: Run the application: python app.py

Step 6: Open a browser and go to: http://localhost:5000

Step 7: Enter Nitrogen, Phosphorous, Potassium, Temperature, Humidity, pH and Rainfall values on the Predict page and click Predict to view the crop recommendation.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Cold-start delay on first request after inactivity	Render's free-tier instance sleeps when idle; first request after a period of inactivity may take 30–60 seconds to respond. Status: Known limitation of free hosting tier.
2	Model accuracy reported as 99.32% on test data	May not generalize to soil/climate conditions outside the training dataset's regional distribution. Status: Documented limitation.
3	No user authentication / saved history	Predictions are stateless; results are not stored per user. Status: Planned enhancement.